OOP Python & Java Interview Handbook

Class

Definition: Class is a core OOP concept. Explain with definition, real-life example, code, and interview points.

Object

Definition: Object is a core OOP concept. Explain with definition, real-life example, code, and interview points.

Constructor

Definition: Constructor is a core OOP concept. Explain with definition, real-life example, code, and interview points.

Inheritance

Definition: Inheritance is a core OOP concept. Explain with definition, real-life example, code, and interview points.

Polymorphism

Definition: Polymorphism is a core OOP concept. Explain with definition, real-life example, code, and interview points.

Encapsulation

Definition: Encapsulation is a core OOP concept. Explain with definition, real-life example, code, and interview points.

Abstraction

Definition: Abstraction is a core OOP concept. Explain with definition, real-life example, code, and interview points.

Python Encapsulation

Single underscore (_var): protected by convention, directly accessible.
Double underscore (__var): name mangling to _ClassName__var, discourages accidental access and avoids inheritance conflicts.

Java vs Python

Java uses compiler-enforced access modifiers (private/public/protected). Python relies on conventions and name mangling.

Top Interview Questions

Q1. What is OOP?
Answer: Prepare a 30-60 second explanation with one example. Key points: definition, purpose, example, and language-specific differences.

Q2. Difference between class and object?
Answer: Prepare a 30-60 second explanation with one example. Key points: definition,

purpose, example, and language-specific differences.

Q3. What is self in Python?
Answer: Prepare a 30-60 second explanation with one example. Key points: definition, purpose, example, and language-specific differences.

Q4. What is a constructor (__init__)?
Answer: Prepare a 30-60 second explanation with one example. Key points: definition, purpose, example, and language-specific differences.

Q5. Types of inheritance?
Answer: Prepare a 30-60 second explanation with one example. Key points: definition, purpose, example, and language-specific differences.

Q6. What is multiple inheritance and MRO?
Answer: Prepare a 30-60 second explanation with one example. Key points: definition, purpose, example, and language-specific differences.

Q7. Method overriding?
Answer: Prepare a 30-60 second explanation with one example. Key points: definition, purpose, example, and language-specific differences.

Q8. Does Python support method overloading?
Answer: Prepare a 30-60 second explanation with one example. Key points: definition, purpose, example, and language-specific differences.

Q9. What is polymorphism?
Answer: Prepare a 30-60 second explanation with one example. Key points: definition, purpose, example, and language-specific differences.

Q10. Explain duck typing.
Answer: Prepare a 30-60 second explanation with one example. Key points: definition, purpose, example, and language-specific differences.

Q11. What is encapsulation?
Answer: Prepare a 30-60 second explanation with one example. Key points: definition, purpose, example, and language-specific differences.

Q12. Difference between _var and __var?
Answer: Prepare a 30-60 second explanation with one example. Key points: definition, purpose, example, and language-specific differences.

Q13. Is __var truly private?
Answer: Prepare a 30-60 second explanation with one example. Key points: definition, purpose, example, and language-specific differences.

Q14. Why use getters/setters?
Answer: Prepare a 30-60 second explanation with one example. Key points: definition, purpose, example, and language-specific differences.

Q15. Difference between encapsulation and abstraction?
Answer: Prepare a 30-60 second explanation with one example. Key points: definition, purpose, example, and language-specific differences.

Q16. Abstract class vs interface (Java)?
Answer: Prepare a 30-60 second explanation with one example. Key points: definition, purpose, example, and language-specific differences.

Q17. What is super()?
Answer: Prepare a 30-60 second explanation with one example. Key points: definition, purpose, example, and language-specific differences.

Q18. What are static/class methods?
Answer: Prepare a 30-60 second explanation with one example. Key points: definition, purpose, example, and language-specific differences.

Best Quick Answers

Encapsulation: Bundle data and methods; Java enforces access with private, Python uses naming conventions and name mangling.
Abstraction: Hide implementation details and expose essential functionality.
Polymorphism: One interface, many implementations.